INCH-POUND

MIL-PRF-83502/2F
3 July 1997
SUPERSEDING
MIL-S-83502/2E
11 December 1989

PERFORMANCE SPECIFICATION SHEET

SOCKETS PLUG-IN ELECTRONIC COMPONENTS, ROUND STYLE (T05, PRESS TO FIT)

This specification is approved for use by all Departments and agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-PRF-83502.

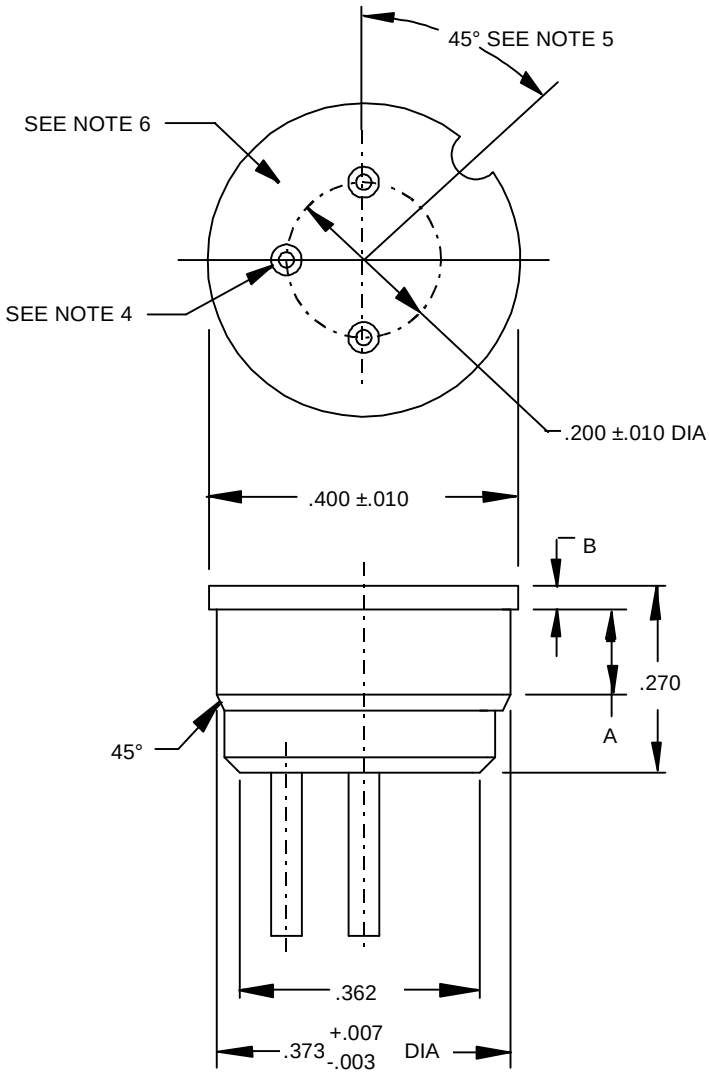
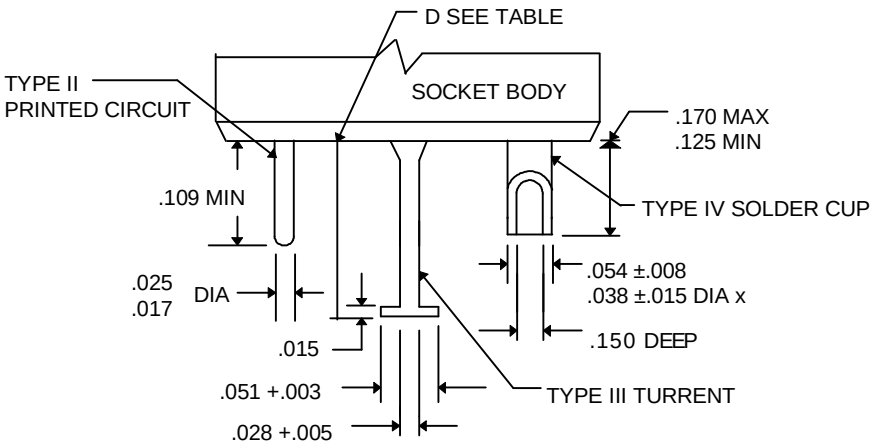


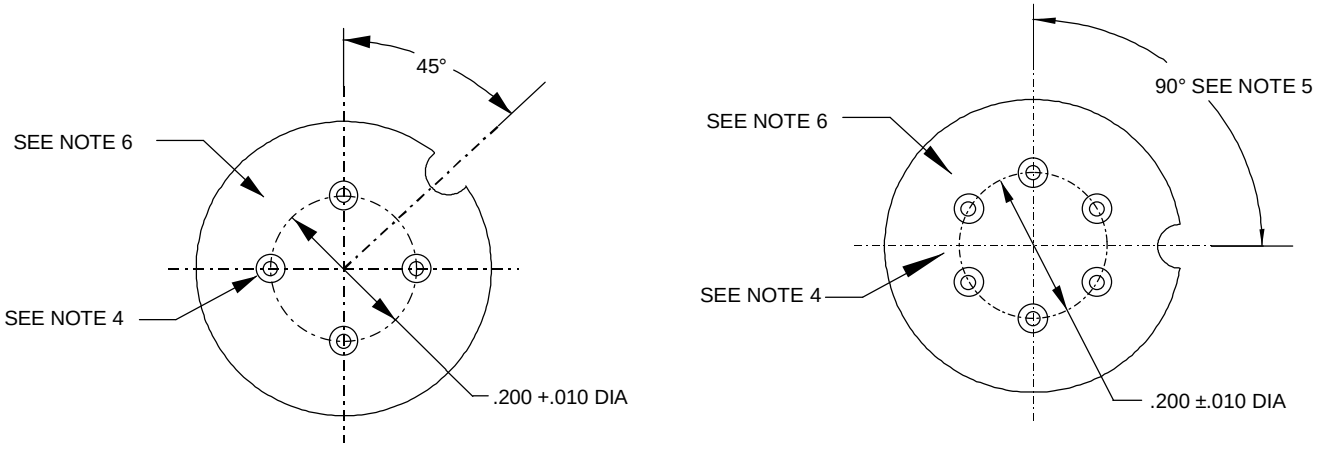
FIGURE 1. Dimensions and configuration.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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TERMINAL DIMENSION

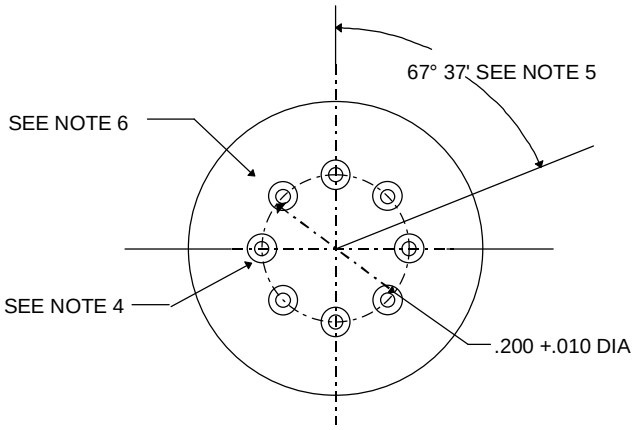


Dash numbers -006, 007, -088

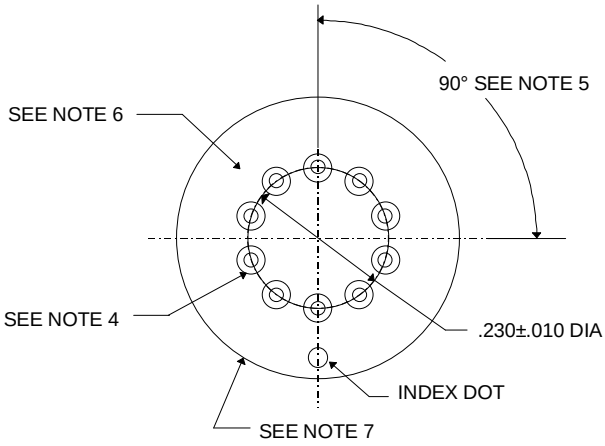
Dash numbers -010, -011, -12

FIGURE 1. Dimensions and configurations - Continued.

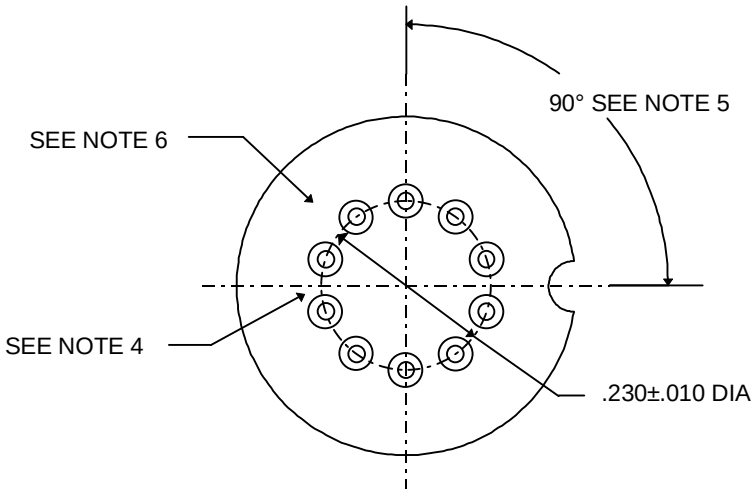
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Dash numbers -014, -015, -016



Dash number 022



Dash numbers -018, -019, -020

FIGURE 1. Dimensions and configurations - Continued.

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Dash number	Termination type	Number of contacts	Dimensions inches (mm)				
			A ±.031 (0.79	B	C ±.010 (0.25)	D ±.015 (0.38)	E ±.003 (0.08)
002	II	3	.150 (3.81)	.060 (1.52)	.270 (6.86)		
003	III	3	.150 (3.81)	.060 (1.52)	.270 (6.86)	.160 (4.06)	.028 (0.71)
004	IV	3	.150 (3.81)	.060 (1.52)	.270 (6.86)		
006	II	4	.150 (3.81)	.060 (1.52)	.270 (6.86)		
007	III	4	.150 (3.81)	.060 (1.52)	.270 (6.86)	.160 (4.06)	.028 (0.71)
008	IV	4	.150 (3.81)	.060 (1.52)	.270 (6.86)		
010	II	6	.150 (3.81)	.060 (1.52)	.270 (6.86)		
011	III	6	.150 (3.81)	.060 (1.52)	.270 (6.86)	.093 (2.36)	.035 (0.89)
012	IV	6	.150 (3.81)	.060 (1.52)	.270 (6.86)		
014	II	8	.150 (3.81)	.060 (1.52)	.270 (6.86)		
015	III	8	.150 (3.81)	.060 (1.52)	.270 (6.86)		
016	IV	8	.150 (3.81)	.060 (1.52)	.270 (6.86)		
018	II	10	.150 (3.81)	.060 (1.52)	.270 (6.86)		
019	III	10	.150 (3.81)	.060 (1.52)	.270 (6.86)	.093 (2.36)	.035 (0.89)
020	IV	10	.150 (3.81)	.060 (1.52)	.270 (6.86)		
022 see note 7	II	10	.150 (3.81)	.060 (1.52)	.270 (6.86)		

FIGURE 1. Dimensions and configuration - Continued.

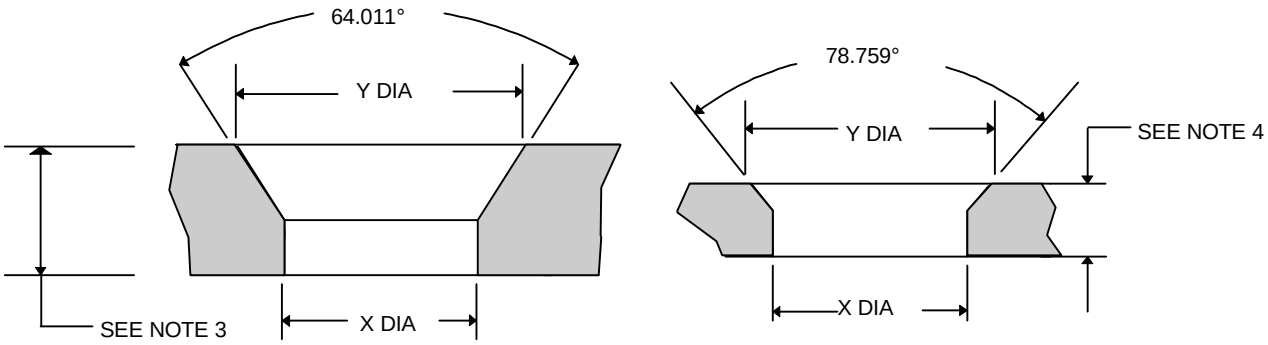
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Inches	mm
.003	0.08
.005	0.13
.006	0.15
.007	0.18
.008	0.20
.010	0.25
.015	0.38
.017	0.4
.025	0.64
.028	0.71
.031	0.89
.035	0.97
.038	1.30
.051	1.37
.054	1.52
.060	2.36
.093	2.77
.109	3.18
.125	3.81
.150	4.06
.160	4.32
.170	5.08
.200	5.84
.230	6.86
.270	9.19
.362	9.47
.373	10.16
.400	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 inch (0.13 mm).
4. Device -002 through -020, to accept leads .017 inch (0.43 mm) diameter + .003 inch (0.08 mm) -.001 inch (0.03 mm) by .156/.218 inch (3.96/5.54 mm) long. Devices -022, .017 inch (0.43 mm) diameter +.003 inch (0.08 mm) -001 inch (0.03 mm) by .125/.165 inch (3.18/4.19 mm). Contact shall be made at or before reaching a depth of .100 inch (2.54).
5. A visual polarization index to be provided in the first quadrant. Location of index dot (inside or outside the socket circle) shall be optional.
6. Marking shall be on the package.
7. Socket is cylindrical in shape having .046 inch (10.31 mm) maximum diameter and is .165 inch (4.19 mm) long, ±0.10 inch (0.25 mm).

FIGURE 1. Dimensions and configurations.

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Busing minor diameter see figure 1		Hole or punch diameter		Diameter "Y"	
Inches +.007 -.003	mm +0.18 -0.08	Inches ±.002	mm ±0.05	Inches +.010 -.000	mm +0.25 -0.00
3.73	9.47	.366/.369	9.30/9.37	.380	9.65

NOTES:
1. For recommended installation procedure drill or punch hole in chassis for press-fit of socket.
2. For printed circuit mounting drill holes to accept terminals.
3. Use 60° countersink for chassis thickness .040 inch (1.02 mm) and over.
4. Use 82° countersink for chassis thickness .039 inch (1.02 mm) and under.
5. Dimensions are in inches.
6. Metric equivalents are given for general information only.
7. Metric equivalents are in parentheses.

FIGURE 2. Recommend mounting.

REQUIREMENTS:

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Dimensions and configurations: See figure 1, and table I.

Reference material: (see 3.1.1 of MIL-PRF-83502).

Body: PTFE (polytetrafluoroethylene) rod in accordance with ASTM D1710.

Contact: Beryllium Copper in accordance with ASTM B139, ASTM B194, ASTM B103, or C729000 spinodal alloy in accordance with ASTM B740.

Contact finish: Gold over nickel in accordance with MIL-PRF-83502.

Terminal: Brass in accordance with ASTM B16.

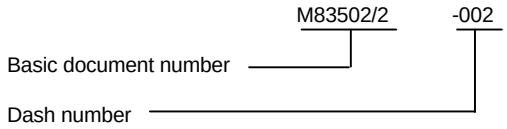
TABLE I. Dash numbers and characteristics.

Dash number M83502/2-	Number of contacts	Termination type	Mating force (maximum) lbs (kg)	Retention force (minimum) lbs (kg)
002	3	II	4.0 (1.81)	4.5 (2.04)
003	3	III	4.0 (1.81)	4.5 (2.04)
004	3	IV	4.0 (1.81)	4.5 (2.04)
006	4	II	4.0 (1.81)	4.5 (2.04)
007	4	III	4.0 (1.81)	4.5 (2.04)
008	4	IV	4.0 (1.81)	4.5 (2.04)
010	6	II	2.5 (1.13)	3.0 (1.36)
011	6	III	2.5 (1.13)	3.0 (1.36)
012	6	IV	2.5 (1.13)	3.0 (1.36)
014	8	II	2.5 (1.13)	3.0 (1.36)
015	8	III	2.5 (1.13)	3.0 (1.36)
016	8	IV	2.5 (1.13)	3.0 (1.36)
018	10	II	2.5 (1.13)	3.0 (1.36)
019	10	III	2.5 (1.13)	3.0 (1.36)
020	10	IV	2.5 (1.13)	3.0 (1.36)
022	10	II	2.5 (1.13)	3.0 (1.36)

These sockets are intended for use on .093 nominal printed circuit boards.

Part or identifying number (PIN): The PIN shall consist of the basic number of this specification sheet and the dash number from table I.

Example



The Government PIN, specified in table III, supersedes the following commercial part numbers.

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TABLE II. Supersession and cross reference data.

Active government PIN	CAGE 91506
M83502/2-002	M8058-1G23
M83502/2-003	M8058-45G1
M83502/2-004	M8058-1G29
M83502/2-006	M8058-1G24
M83502/2-007	M8058-45G2
M83502/2-008	M8058-1G30
M83502/2-010	M8058-1G33
M83502/2-011	M8058-1G39
M83502/2-012	M8058-1G18
M83502/2-014	M8058-1G32
M83502/2-015	M8058-1G37
M83502/2-016	M8058-1G19
M83502/2-018	M8058-1G31
M83502/2-019	M8058-1G38
M83503/2-020	M8058-1G22
M83502/2-022	M8058-1G91

NOTE: Revision letters are not used due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodian:
Navy - EC
ARMY - CR
Air Force - 85

Preparing activity:
DLA - CC

(Project 5935 - 4034-02)

Review activity:
Navy - AS, MC, SH
Air Force - 11, 19, 99
Army - AR, AT, AV, MI, ME

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